

Introduction to Machine Learning

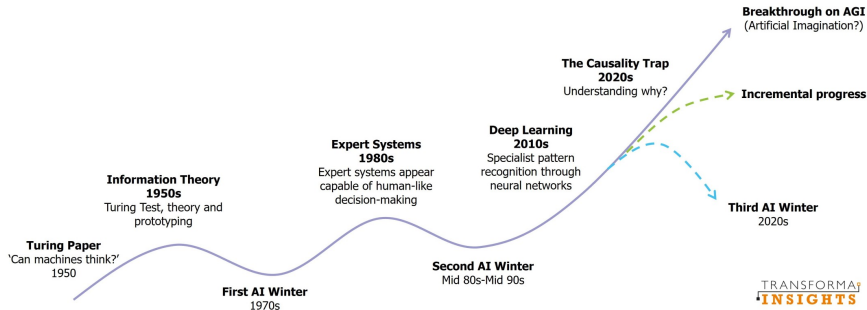
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- 1 AI - Historical perspective
- 2 AI's invisible hand in daily life
- 3 AI's invisible hand in Medicine?
- 4 Introduction to Machine Learning

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AI - Historical perspective



Why AI boomed in the 2010s

Increased computational power

- Growth in GPU and TPU capabilities



- Cloud infrastructures

Big data explosion

- Vast amounts of data available from social media, IoT, and online platforms

Advances in Machine Learning algorithms

- Rise of deep learning and neural networks (e.g., AlexNet 2012 breakthrough)

Open source ecosystem

- Platforms like TensorFlow and PyTorch made AI tools more accessible

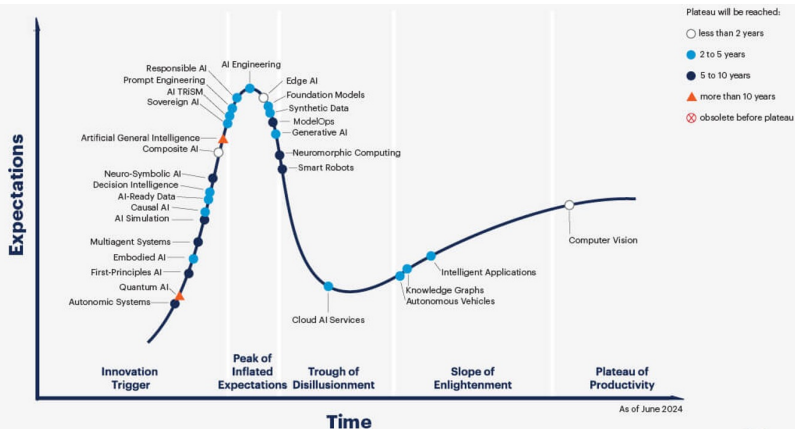
Investment and industry adoption

- Significant investments by tech giants (Google, Amazon, etc.)

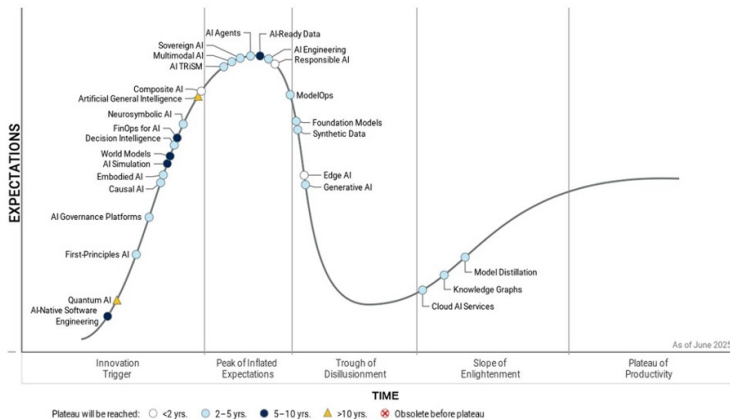
Hype Cycle for Artificial Intelligence, 2023



Hype cycle for AI, 2024



Hype cycle for AI, 2025



Gartner

A critical look at the "Publish or Perish" culture

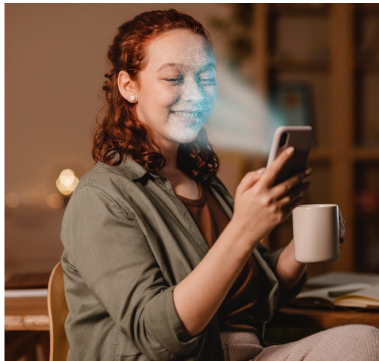


Summary

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Open your phone with face ID

- Open your phone with face ID (pattern recognition)



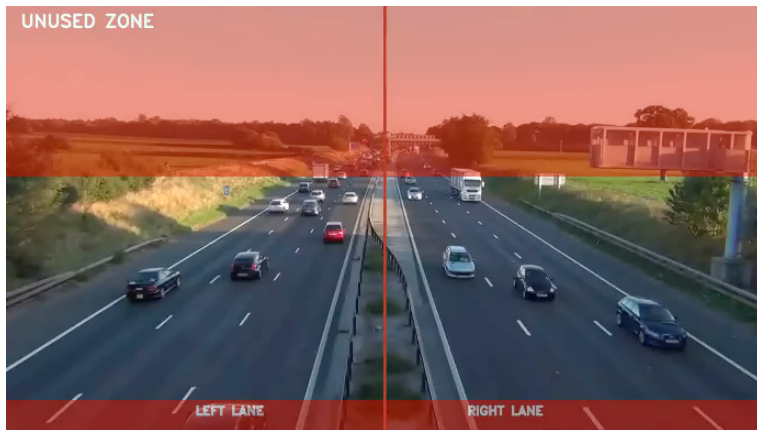
Streaming services

- Streaming services: Use past viewing history to deliver suggestions for what you might want to watch (prediction)



Traffic monitoring

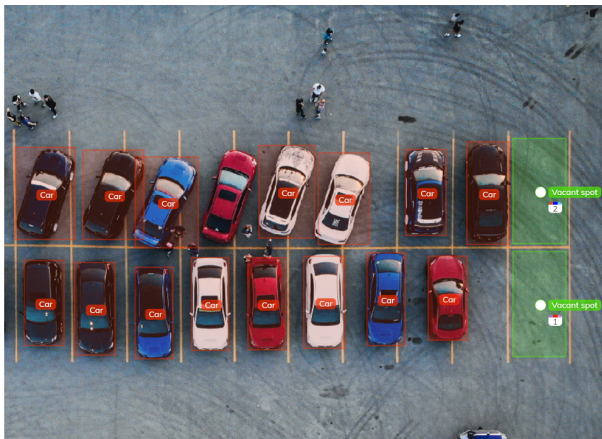
- Traffic monitoring (detection)



Play Demo Video

Parking occupancy detection

- Parking occupancy detection: it is an alternative to more costly sensor-based technologies that require regular maintenance (detection)



Road maintenance

- Road maintenance: Increasing road maintenance allocation efficiency and decreasing the safety risk related to accidents



Play Demo Video

Road maintenance

- Road maintainance: Increasing road maintenance allocation efficiency and decreasing the safety risk related to accidents



Figure: Robotiz3d and University of Liverpool

Summary

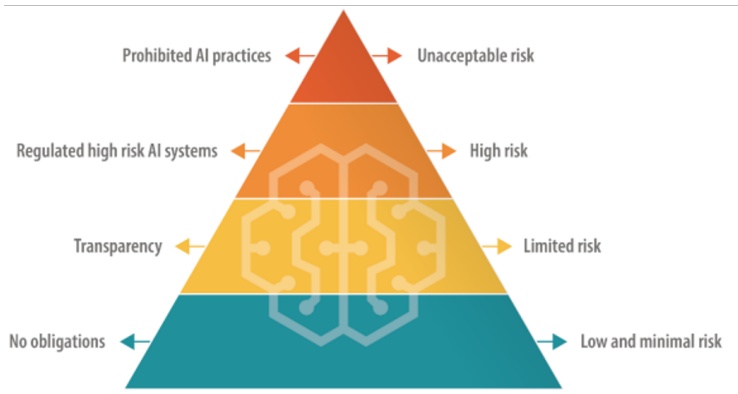
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Is it always a good thing that AI works as an invisible hand?

E.g., in high-risk applications like medicine?

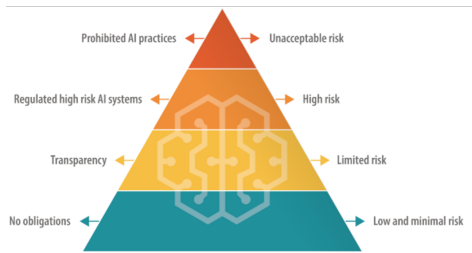
European AI act

Regulatory framework proposal on artificial intelligence



European AI act

Regulatory framework proposal on artificial intelligence

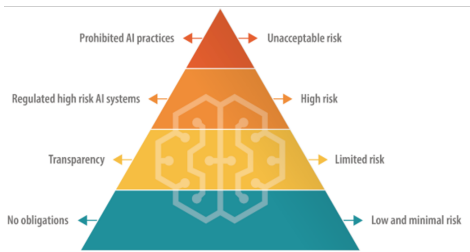


Unacceptable risk: clear threat to the safety, livelihoods, and rights of people — these systems will be banned.

Examples: social scoring by governments, toys using voice assistance that encourages dangerous behavior.

European AI act

Regulatory framework proposal on artificial intelligence

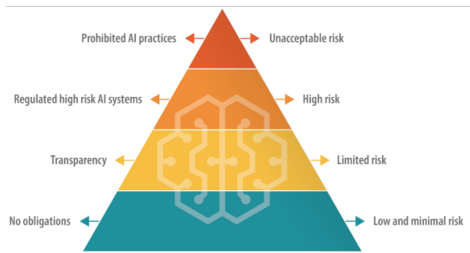


High risk:

- critical infrastructures (e.g., transport), that could put the life and health of citizens at risk;
- safety components of products (e.g., AI application in robot-assisted surgery);
- migration, asylum , and border control management (e.g., verification of authenticity of travel documents);

European AI act

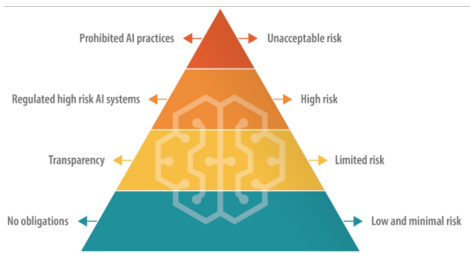
Regulatory framework proposal on artificial intelligence



Limited risk: when using AI systems such as chatbots, users should be aware that they are interacting with a machine so they can take an informed decision to continue or step back.

European AI act

Regulatory framework proposal on artificial intelligence



Low and minimal risk: free use of minimal-risk AI. This includes applications such as AI-enabled video games or spam filters.

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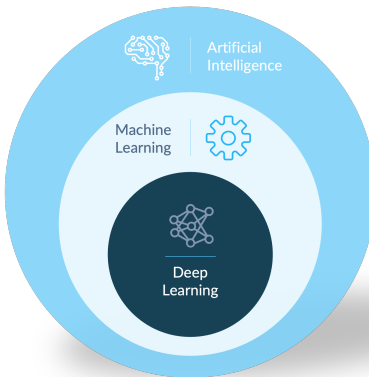
What is AI?

Artificial Intelligence

The theory and development of computer systems able to perform tasks normally requiring human intelligence

Deep Learning

Subset of ML techniques which use complex algorithms (artificial neural networks) to learn from large amounts of data



Machine Learning

Subset of AI techniques which use statistical learning to enable machines to improve with experience

What is machine learning?

“Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.”

— Arthur Samuel, 1958



Arthur Samuel (1901–1990)

Example of explicit programming

Traditional approach – pseudo-code algorithm

Example: Maximum of Two Values

Algorithm 1 Find the Maximum of Two Values

```
1: Input:  $a, b$   
2: if  $a > b$  then  
3:    $max \leftarrow a$   
4: else  
5:    $max \leftarrow b$   
6: end if  
7: Output:  $max$ 
```

**However, for some tasks, we do not have
an algorithm!**

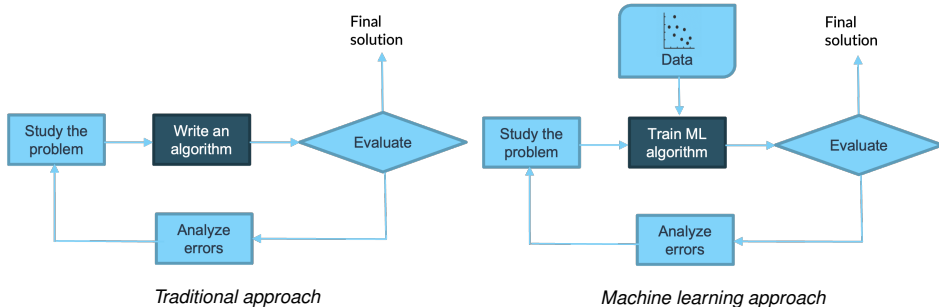


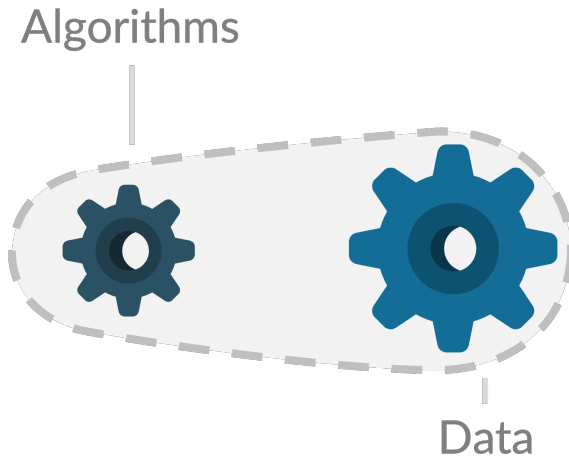
Face recognition:

- Humans do it unconsciously and are unable to explain how they do it
- No explanation, no traditional algorithm
- A person's face is a pattern
- A learning program captures the pattern specific to that person

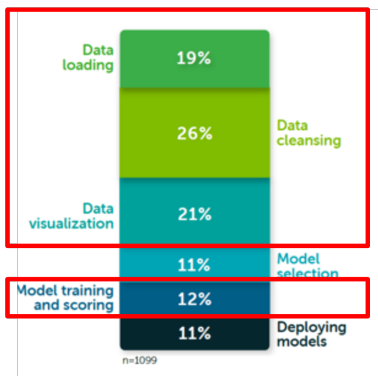
Machine learning approach

What we lack in knowledge, we make up for in data





Data preparation in Machine learning



- The 2020 State of Data Science is based on online surveys of nearly 2,400 people from more than 100 countries (not all of whom are data scientists themselves, although they work in the field)
- Data preparation is a highly time-intensive part of any machine learning project
- It could be worse. In some surveys in the past, data prep tasks have occupied upwards of 70% to 80% of a data scientist's time

2020 State of Data Science: Moving From Hype Toward Maturity.

Machine learning - The big picture



Genetic analysis



Multimodal neuroimaging data



Test scores



Clinical examination



Lab analysis



or

Machine Learning algorithm



E.g., patients vs. healthy controls classification

Data

Information

Machine learning - The big picture



Genetic analysis



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Machine Learning algorithm

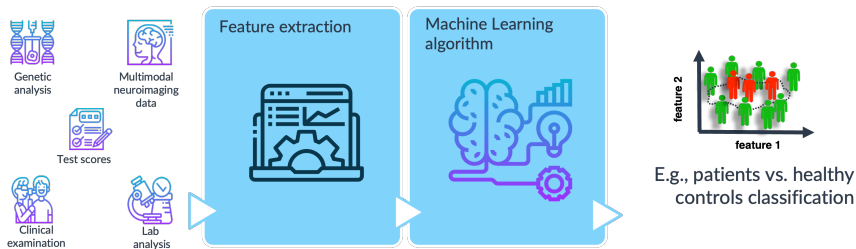


E.g., patients vs. healthy controls classification

Data

Information

Machine learning - The big picture



Data

Information

Deep learning - The big picture

